

Study guide for Exam DP-900: Microsoft Azure Data Fundamentals

Purpose of this document

This study guide should help you understand what to expect on the exam and includes a summary of the topics the exam might cover and links to additional resources. The information and materials in this document should help you focus your studies as you prepare for the exam.

 Expand table

Useful links	Description
How to earn the certification	Some certifications only require passing one exam, while others require passing multiple exams.
Your Microsoft Learn profile	Connecting your certification profile to Microsoft Learn allows you to schedule and renew exams and share and print certificates.
Exam scoring and score reports	A score of 700 or greater is required to pass.
Exam sandbox	You can explore the exam environment by visiting our exam sandbox.
Request accommodations	If you use assistive devices, require extra time, or need modification to any part of the exam experience, you can request an accommodation.
Take a free Practice Assessment	Test your skills with practice questions to help you prepare for the exam.

Updates to the exam

Our exams are updated periodically to reflect skills that are required to perform a role. We have included two versions of the Skills Measured objectives depending on when you are taking the exam.

We always update the English language version of the exam first. Some exams are localized into other languages, and those are updated approximately eight weeks after the English version is updated. Although Microsoft makes every effort to update localized versions as noted, there may be times when the localized versions of an exam are not updated on this schedule. Other available

languages are listed in the **Schedule Exam** section of the **Exam Details** webpage. If the exam isn't available in your preferred language, you can request an additional 30 minutes to complete the exam.

Note

The bullets that follow each of the skills measured are intended to illustrate how we are assessing that skill. Related topics may be covered in the exam.

Note

Most questions cover features that are general availability (GA). The exam may contain questions on Preview features if those features are commonly used.

Skills measured as of November 1, 2024

Audience profile

This exam is an opportunity to demonstrate your knowledge of core data concepts and related Microsoft Azure data services. As a candidate for this exam, you should have familiarity with Exam DP-900's self-paced or instructor-led learning material.

This exam is intended for you, if you're a candidate beginning to work with data in the cloud.

You should be familiar with:

- The concepts of relational and non-relational data.
- Different types of data workloads such as transactional or analytical.

You can use Azure Data Fundamentals to prepare for other Azure role-based certifications like Azure Database Administrator Associate or Azure Data Engineer Associate, but it is not a prerequisite for any of them.

Skills at a glance

- Describe core data concepts (25–30%)
- Identify considerations for relational data on Azure (20–25%)

- Describe considerations for working with non-relational data on Azure (15–20%)
- Describe an analytics workload on Azure (25–30%)

Describe core data concepts (25–30%)

Describe ways to represent data

- Describe features of structured data
- Describe features of semi-structured
- Describe features of unstructured data

Identify options for data storage

- Describe common formats for data files
- Describe types of databases

Describe common data workloads

- Describe features of transactional workloads
- Describe features of analytical workloads

Identify roles and responsibilities for data workloads

- Describe responsibilities for database administrators
- Describe responsibilities for data engineers
- Describe responsibilities for data analysts

Identify considerations for relational data on Azure (20–25%)

Describe relational concepts

- Identify features of relational data

- Describe normalization and why it is used
- Identify common structured query language (SQL) statements
- Identify common database objects

Describe relational Azure data services

- Describe the Azure SQL family of products including Azure SQL Database, Azure SQL Managed Instance, and SQL Server on Azure Virtual Machines
- Identify Azure database services for open-source database systems

Describe considerations for working with non-relational data on Azure (15–20%)

Describe capabilities of Azure storage

- Describe Azure Blob storage
- Describe Azure File storage
- Describe Azure Table storage

Describe capabilities and features of Azure Cosmos DB

- Identify use cases for Azure Cosmos DB
- Describe Azure Cosmos DB APIs

Describe an analytics workload (25–30%)

Describe common elements of large-scale analytics

- Describe considerations for data ingestion and processing
- Describe options for analytical data stores
- Describe Microsoft cloud services for large-scale analytics, including Azure Databricks and Microsoft Fabric

Describe consideration for real-time data analytics

- Describe the difference between batch and streaming data
- Identify Microsoft cloud services for real-time analytics

Describe data visualization in Microsoft Power BI

- Identify capabilities of Power BI
- Describe features of data models in Power BI
- Identify appropriate visualizations for data

Study resources

We recommend that you train and get hands-on experience before you take the exam. We offer self-study options and classroom training as well as links to documentation, community sites, and videos.

 Expand table

Study resources	Links to learning and documentation
Get trained	Choose from self-paced learning paths and modules or take an instructor-led course
Find documentation	Azure SQL documentation - Azure SQL SQL Server technical documentation - SQL Server Azure Blob Storage documentation Azure Table storage documentation Azure Storage documentation Azure Cosmos DB Azure Synapse Analytics Azure Databricks Data Factory Power BI documentation - Power BI
Ask a question	Microsoft Q&A Microsoft Docs

Study resources	Links to learning and documentation
Get community support	Analytics on Azure TechCommunity Azure Synapse Analytics TechCommunity Welcome to the SQL Server Community (microsoft.com) Azure PaaS - Microsoft Tech Community
Follow Microsoft Learn	Microsoft Learn - Microsoft Tech Community
Find a video	Exam Readiness Zone Data Exposed Browse other Microsoft Learn shows

Change log

Key to understanding the table: The topic groups (also known as functional groups) are in bold typeface followed by the objectives within each group. The table is a comparison between the two versions of the exam skills measured and the third column describes the extent of the changes.

 Expand table

Skill area prior to November 1, 2024	Skill area as of November 1, 2024	Change
Describe an analytics workload on Azure	Describe an analytics workload	Minor
Describe common elements of large-scale analytics	Describe common elements of large-scale analytics	Minor

Last updated on 11/02/2024